How can libraries do respectful requirements elicitation in an Indigenous Data and AI Context?

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Introduction

This paper presents the findings of the AHRC-funded iREAL (Inclusive Requirements Elicitation for AI in Libraries) project, which aimed to develop a model for responsible AI systems development in libraries seeking to include knowledge from Indigenous communities, specifically Aboriginal and Torres Strait Islander communities in Australia. As Indigenous peoples continue to advocate for their rights and wellbeing, including sovereignty over their data (Carroll *et al.*, 2020), how should Research Technology Professionals (RTPs) elicit requirements for AI/Machine Learning projects incorporating Indigenous data in library collections?

Research Context

Globally, libraries hold collections extracted from Indigenous communities, often collected under colonial missions. However, there is a tension between the current trajectory of AI development as a Western rationalist endeavour, and the multiple ways of knowing within Indigenous Knowledge systems (Lewis, Whaanga and Yolgörmez, 2024). While Indigenous researchers have proposed guidelines for Indigenous Data Governance (IDGov) (Maiam nayri Wingara, 2018)that can support and promote Indigenous Data Sovereignty (IDSov) (Cummins *et al.*, 2023), there has been little attempt to understand how such principles might inform AI systems development and critique within libraries.

iREAL therefore aimed to develop guidance on responsible AI systems development in libraries that embeds Indigenous rights and perspectives into the principles of "respect for human autonomy, prevention of harm, fairness and explicability" that are the ethical basis of trustworthy AI (High Level Expert Group on Artificial Intelligence, 2019). The project brought together Indigenous and non-Indigenous researchers from the University of Glasgow, King's College London, and University of Technology Sydney, to address two key objectives that will be covered in this presentation:

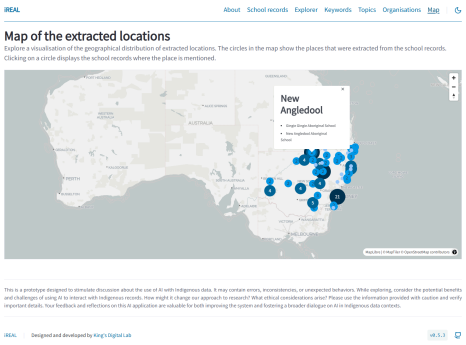
1. Understanding the application of Indigenous rights and IDGov to library AI systems in Australia and the United Kingdom.
2. Propose a preliminary model for inclusive requirements elicitation in library AI system assessment or development.

Methodology

We utilised a workshop-based approach, inviting researchers, library practitioners and RTPs to analyse how responsible AI systems might be critiqued and created including the knowledge and data of Indigenous communities. Our methodology drew on three key texts: Linda Tuhiwai Smith's *Decolonizing Methodologies: Research and Indigenous Peoples* (2021), which encourages project participants to be proactive and interested workers for change to address structural and cultural reform; the *Indigenous Archives Collective Position Statement on the Right to Reply to Indigenous Knowledges and Information Held in Archives* (2021), which emphasises a "participatory model that repositions record 'subjects', individuals and communities as record 'agents' and participants in the act of record creation"; and the Maiam nayri Wingara Collective Principles of IDSov (2018) that refer to the right of Indigenous peoples to autonomously decide what, how and why Indigenous data are collected, accessed and used.

The first workshop, at the National Library of Australia, was attended by approximately 35 researchers and library practitioners from a majority Indigenous Australian background. Taking place over a day, a series of Yarning sessions encouraged participants to address contexts and challenges relating to AI development in the Australian context. These findings formed the basis of planning for workshop 2, which took place at King's

College London with approximately 20 attendees. Led by King's Digital Lab, it explored the legacies and challenges of applying AI to Indigenous datasets. In advance, KDL prepared a dashboard that applied a Large Language Model to historical records about New South Wales Aboriginal Schools 1876-1979, from Museums of History New South Wales. These records were converted to JSON and Markdown, and AI/ML models for metadata extraction were then applied. Finally, a prototype dashboard was produced which showcased the results including basic topic modelling, named entity recognition, and mapping. Figure 1 below shows the dashboard mapping interface. This dashboard formed the basis of a critical investigation into the outputs of the LLM. Groups collaboratively investigated the dashboard, before reverting to larger Yarning circles to discuss their findings. The workshop finished with a requirements elicitation session, that drew together the day's findings into a focused discussion on notional priorities for future development. This was done via a MoSCoW (Must-Have, Should-Have, Could-Have, Won't Have) prioritisation exercise.



Screenshot of the iREAL dashboard, used for Workshop 2

Finally, in workshop 3, which was online with a smaller group of 15 attendees from the previous two workshops, we undertook a critical investigation of how Retrieval Augmented Generation (RAG) worked on the same dataset. This final workshop finished with discussion of the gaps in our discussion, and identified issues and challenges associated with Requirements Elicitation.

We encountered significant time pressures due to the six-month duration of the project. This had a significant effect on our relationship building, in selecting a dataset, and in planning engagement activities. However, it also led to the development of an effective reflective approach to analysing our research findings. As well as analysing contributions from workshop participants, the project team also shared short written reflections about each workshop. This became an emergent intra-project information sharing technique that is evocative of auto-ethnographic practices: an "autobiographical genre of writing and research that displays multiple layers of consciousness, connecting the personal to the cultural" (Bochner and Ellis, 2016, p. 65). These reflections form the basis of our findings and inform the subsequent recommendations that will be outlined in the presentation.

Findings

A key learning outcome for the team was the importance of developing a suitable cultural interface, where "the intersections of different Knowledges, systems, concerns and priorities will converge to inform and develop new practices in this area" (Nakata, 2002). The development of a respectful requirements elicitation process therefore must focus on how it navigates different Knowledge Structures, priorities, systems and concerns in order to develop effective cross-cultural spaces for investigating AI usage within libraries. The question of who is in the room was key. This operated on two levels: first, to ensure that communities were adequately represented in discussions; and second, in response to concerns from participants about the importance of understanding the contexts in which this work takes place. Many problems identified with the Workshop 2 dashboard, for instance, were found to originate in a lack of knowledge of the histories, politics and cultures that produced the datasets and technologies that were under critique.

Indigenous relationality – an acknowledgment of the multiple and complex relationships that humans have with each other and the natural world – was a common theme. This encompassed a variety of challenges: the need to engage with communities in their own places and spaces; the question of whether AI should be used in certain contexts, given its environmental impact; the desire among participants to actively dismantle colonial practices; This the need for an awareness of Indigenous concerns and imperatives in the AI space, and how these relate to the collections and practices of libraries; the need for transparency at all levels in selecting methodologies, datasets, and AI technologies; and the moments in which the Right to Reply is invited or signposted. Importantly, it was evident from our attempt to utilise Indigenous methodologies such as Yarning in very different cultural contexts, that a degree of understanding of the social and cultural roots of the methods being adopted was necessary and that

context was a key driver for how to engage with communities.

This was summarised neatly by participant calls for a “Slow AI” approach in the sector – not one built upon inaction, but upon knowledge exchange and reflection to address key concerns and seek pathways of mutual benefit to institutions, RTPs and Indigenous communities alike. Furthermore, the deployment of relational methods is highly context specific and thus challenges the cultural hegemony of many existing AI practices. Any sectoral response must therefore be simultaneously localised and built upon global foundations of Indigenous leadership, and Indigenous Data Governance.

Discussion and Conclusion

From these findings have emerged a set of recommendations that can support RTPs to undertake respectful AI systems development for Indigenous knowledges in libraries. In addition to supporting the adoption of existing frameworks for IDSov and IDGov, our recommendations emerge from the key workshop themes of relationality, relationship-building, and Indigenous representation. As such, the recommendations we will outline in our presentation include: allowing time, and creating inclusive spaces for, relationship building among project teams and within communities; integrating RTPs into leadership roles, teams and processes; developing and adapting RTP practices for the Indigenous contexts in which this work will take place; engaging early with data and use cases, exploring their formats and possible outputs, and drawing on team knowledge domain expertise; foregrounding transparency by explaining why datasets and AI models are selected, sharing process documentation, owning mistakes and learning from failures; and strongly aligning RTP practices towards collaborative change, interventions and advocacy for Indigenous rights.

This paper aims to make a significant contribution to our understanding of Responsible AI practices for RTPs and libraries, by intervening in development practices relating to the collections and knowledges of Indigenous communities. While grounded in the specific nationhood of Aboriginal and Torres Strait Islander peoples, the presentation therefore has wider significance for those facing similar challenges in their own localised contexts.

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